

Knowledge ledger — barehands, anatomy edition

What this fork has **learned, decided, and verified**, with the evidence for each line. Code shows *what* the project does; this file records *why*, and which lessons cost a field report to learn. Read it before changing the medical layer, so you don't re-learn something already paid for.

How to use it

- One entry per fact. Each entry names its evidence: a commit, a file, or a test run. If something has no evidence, it goes under *Open items*, not in the ledger.
- Tag every entry: **LAW** (a rule you don't break), **FACT** (measured or verified), **LESSON** (a bug, what caused it, and the fix), **OPEN** (not known yet).
- Add new entries; don't rewrite old ones. When something stops being true, strike it through (*~~...~~*) and add the entry that replaces it, so the history stays readable.
- Add dated lines to the *Log* at the bottom.

1. Status snapshot — 2026-09-16

thing	state	evidence
Contract suite	77/77 pass	python3 tests/test_contract.py , run 2026-09-16
Gesture suite	54/54 pass	node tests/test_gesture.mjs , run 2026-09-16
Heart sidecar ↔ GLB	valid, 0 warnings, 16/16 meshes covered	schemas/validate.py , run 2026-09-16
Anterior MI case	valid, 0 warnings; review.status: verified	validate.py --case , run 2026-09-16; 36caa25
Visual suite (17 renders)	not re-run since 2f7df88	tests/test_visual.py takes ~5 min
Upstream	1 commit not merged (aa0e860 , README only)	git rev-list main...upstream/main
Organs	1 (heart)	media/models/heart/
Cases	1 (heart-anterior-stemi-01)	cases/

2. Laws — rules you don't break

- **LAW — Education only.** This is not a medical device. It doesn't diagnose real patients, and nothing it shows is clinical judgment about a real person. CLAUDE.md , README.md
- **LAW — Content is never invented at runtime.** Every structure name, teaching line and pathology claim is read from a sidecar or case that a human wrote and a reviewer checked. narrate.sh speaks only sidecar text; no model runs in that loop. 6dd96ab
- **LAW — The media airlock never widens.** The board stages files only from media/ . A bare organ basename (heart.glb) gets a 400 on purpose: every organ exists in both models/ and holo/ , so the name is ambiguous and the server refuses to guess. 2f7df88 , tests/test_contract.py
- **LAW — The /cmd allowlist grows only through validated verbs.** Each new verb gets strict stdlib validation (enums, ranges, id regexes); no bare passthrough. 5af0f6a , schemas/verbs.md

- **LAW — Answer isolation.** `/case-render` strips the vignette, diagnosis and teaching. Tests grep the response for the answer *text*, not only the field names. The examiner must never `label` the structure a lesion sits on, because the label gives away the answer. `67d2c84`, `2f7df88`
- **LAW — A case runs only once `review.status` is verified.** The examiner refuses anything else. `67d2c84`
- **LAW — Fetch a source before writing any reference, including a correction.** Fixing a citation is exactly when fabrication happens: an identifier supplied by the reviewer makes the fix *feel* verified while the byline is invented. `06ab91c`, `36caa25`
- **LAW — `stage.html` changes only through marked `MEDICAL HOOK` lines.** Everything else goes in `js/medical/`, so upstream's gesture fixes merge cleanly. `adac0a6`
- **LAW — A medical hook must never be able to kill hand tracking.** Every call into `MED` is an optional call (`MED.fn?.()`). If a module is stale or half-loaded, the board degrades to plain upstream barehands instead of freezing. `be028b1`
- **LAW — Updaters never merge.** `update.sh` / `update.bat` only fetch upstream and print a collision forecast. You then merge by hand with `git merge upstream/main`. `673bba6`

3. Facts — measured or verified

The heart asset

- **FACT** — `heart.glb` has 16 named meshes built from DBCLS BodyParts3D (CC-BY 4.0), decimated to about 130k tris / 1.5 MB. Attribution is in `media/models/heart/LICENSE.md`. `8f07afa`
- **FACT** — Structure ids: `right_atrium`, `left_atrium`, `right_ventricle`, `left_ventricle`, `ascending_aorta`, `aortic_arch`, `pulmonary_trunk`, `superior_vena_cava`, `inferior_vena_cava`, `right_coronary_artery`, `left_coronary_artery`, `coronary_sinus`, `tricuspid_valve`, `mitral_valve`, `aortic_valve`, `pulmonary_valve`. Valves are on layer 1 and parented to their ventricles; the sidecar has 30 clinical aliases. `fdeb348`
- **FACT** — The heart has **no LAD mesh**. The alias “LAD” maps to the undivided `left_coronary_artery`. `fdeb348`
- **FACT** — The heart's own axes: superior **+z**, left **+x**, anterior **-y**. This is not glTF Y-up, and it is recorded in the sidecar's `axes` block. `d63bef4`
- **FACT** — The GLB has POSITION data only, with **no NORMAL**. `core.js` computes vertex normals at load for any medical model that lacks them. `2f7df88`
- **FACT** — All 16 meshes are closed manifolds, so stencil caps count exactly on this model. Leaky models need a fallback, which is documented in a comment but not built. `4c121d5`

The runtime

- **FACT** — A connected tracker posts `/state` about 45x/sec, so an injected synthetic state lasts about 22 ms. You can't simulate a pointing hit while the board is open. Board-state tests report SKIPPED instead of failing in that case. An idle but live tracker posts byte-identical heartbeats, so matching hashes don't mean the page has stopped. `6dd96ab`, `2a2664e`
- **FACT** — The `/state` heartbeat budget is 256 KB, which is why the bus field names are short (`cut`, `iso`, `hid`, `lb`, `pt`). `schemas/verbs.md`
- **FACT** — The tracker page can't boot headless because MediaPipe needs a real camera. Only a person at the webcam can verify pointing and gesture thresholds. `tests/README.md`
- **FACT** — Render tests need hardware ANGLE. SwiftShader can't compile the cut's stencil programs within the time budget and quietly renders an uncut model. `tests/README.md`
- **FACT** — Tracker hotkeys: **R** re-read notes · **C** cycle camera · **D** debug overlay · **P** pose sampler · **X** arm or disarm the cut (medical) · **G** cut-gesture sampler (medical, `gesture.js`).
- **FACT** — The flat-palm incision **works on a real hand**. This was the first live confirmation of any cut threshold. `879e36d`

Content

- **FACT** — Under the Fourth Universal Definition, a 48 h infarct is **acute**. “Subacute” means the 7–28 day window, and “evolving” means under 6 h. Neither is an accepted answer for the 48 h case. `7c0fbba`

- **FACT** — A flat tint render **can’t** show the two-zone gross appearance (yellow-tan centre with a congested rim). Rubrics may reward only what the board can actually display. `ef2ae07` , `7c0fbba`
- **FACT** — Correct citation for NBK537228: *Lateral Wall Myocardial Infarction*, Ludhwani, Chhabra, Goyal, Quintanilla Rodriguez, Gupta (updated 29 Oct 2024). An earlier byline had been invented. `06ab91c`

4. Lessons — bug → cause → fix

#	symptom	real cause	fix	commit
L1	X-ray rendered nothing (zero non-white pixels)	GLB had no normals, so the shader computed <code>normalize(vec3(0)) = NaN</code>	compute vertex normals at load	<code>2f7df88</code>
L2	“Label all” pushed labels off the card	de-collision pushed labels only downward, with no bound	walk back up and clamp to the card	<code>2f7df88</code>
L3	Coronal cut at t=0.55 missed the myocardium	presets assumed glTF Y-up	anatomical axes block in the sidecar	<code>d63bef4</code>
L4	Heart slowly rotated on its own, and pointing was flaky	upstream’s +0.12 rad/s idle drift is right for a prop, wrong for a specimen	no drift for models with a sidecar; hand spin still coasts	<code>b26172e</code>
L5	Edits to <code>core.js</code> “didn’t take”	browser cached the module	<code>js/medical/*</code> served no-store	<code>b26172e</code>
L6	Tracker froze: <code>MED.stillModel</code> is not a function	stale module plus a throw inside the rAF loop	all hooks became optional calls	<code>be028b1</code>
L7	With a cut armed, every pinch triggered narration	pointing read the <i>suspended</i> pinch flag	read the physical pinch before suspension	<code>2a2664e</code>
L8	Cut mode disarmed before the first swipe	20 s idle timeout was a guess	3 minutes	<code>8693af1</code>
L9	Arming needed an agent round-trip mid-gesture	no local control	X key. A pose was rejected because a flat palm is also the pointing pose	<code>879e36d</code>
L10	“It’s not cutting” on a large heart	swipe travel scaled with card width (6× → 1584 px, wider than the screen)	clamp travel to a span a shoulder can cross (6× → 660 px)	<code>bbf3e94</code>
L11	Pinching near the heart grabbed it	hit area was the fixed 480 px card, which grows with scale	hit-test the organ’s projected silhouette	<code>f0b1c37</code>
L12	Narrated everything the hand swept past	gate checked “same structure 150 ms”, not	26 px stillness + 600 ms rest on the bus;	<code>37d8ff3</code>

		"hand stopped"	narrate.sh --dwell 2 s	
L13	A scrub's release made a cut	an opening hand still moving sideways is a flat palm crossing the organ	reject incisions within 800 ms of a pinch (upstream's clap idiom)	256d792
L14	Flat "cut" faces while exploding	not the cut: parts flew past the fixed camera frame, and cropped open shells show their insides	camera dollies back as the explosion opens	5815850
L15	Armed cut + two flat hands could clear the board	the clap gate screened only recent pinches	an armed cut disqualifies the clap	a50a76f

Meta-lessons - LESSON — Check your diagnosis before building a fix. L13 fixed a real bug, but it wasn't what the user's photo showed (L14). 5815850 - **LESSON** — Comments can say the opposite of what the code does: L10's comment called the travel rule "scale-proof". Measure at the extremes (large scale, small window). bbf3e94 - **LESSON** — In gesture tests, identical poses on identical paths write identical planes, so "the value changed" can't detect a cut. Also, the previous commit's cooldown may still be running. Both are noted in the test. 256d792 - **LESSON** — Independent review catches what the author can't: two content errors (ef2ae07) and one invented citation (06ab91c). Keep reviewers in a separate context, with no write access to the content.

5. Content ledger

id	organ	pathology doc	case	review	notes
heart-anterior-stemi-01	heart	pathology/anterior-mi.md	cases/heart-anterior-stemi-01.json	verified 2026-08-20, 3 passes	48 h timepoint; 22 accepted phrasings; 4 distractors; review files in review/

6. Open items

- **OPEN** — Cut-gesture thresholds are mostly still provisional. The swipe fires on a real hand (879e36d), but palm-normal stability is flagged as the likeliest wrong value and the two-finger retraction scrub has no live confirmation. Run skills/gesture-tuning/SKILL.md .
- **OPEN** — Pointing handedness hasn't been checked by a human at the webcam: does POINTING AT ever name the mirror-image structure? core.js has a marked one-line fix ready. tests/README.md
- **OPEN** — The MI morphology table was confirmed against four independent references, but **not** against the printed Robbins 10e it cites. It stays flagged in the pathology doc. 36caa25
- **OPEN** — Upstream aa0e860 (README example vault path) isn't merged. It touches README.md , which this fork has rewritten, so expect a conflict and keep the fork's version.

- **OPEN** — The `layer` verb has little to show on the heart (layers 0–1 only). `verbs.md` examples use ids that don't exist here (`skin`, `cardiomyopathy-1`); they're only illustrations.
 - **OPEN** — Stencil caps are skipped when post-processing (bloom) is on, because the bloom render targets have no stencil buffer. `4c121d5`
 - **OPEN** — Leaky (non-manifold) models have no solid-ink cap fallback yet. `4c121d5`
 - **OPEN** — The visual suite hasn't been re-run since `b26172e` onward changed render/camera behaviour (drift removed, explode dolly added).
 - **OPEN** — `Claude.pdf` in the repo root is untracked and unreviewed.
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7. Log

- **2026-08-15** → **08-19** — Upstream barehands, 24 commits (`b01a35b ... f1f8c9e`).
- **2026-08-20** — Fork groundwork, heart model and sidecar, medical layer (9 hooks), capped cuts, anatomical axes, hand-driven cut, anterior MI case through three review passes → verified.
- **2026-08-21** — Fork-safe updater, examiner skill, acceptance suite (caught L1/L2), a day of field fixes (L4–L12), `point-watch.sh`, `narrate.sh`.
- **2026-08-22** — Scrub release guard (L13), explode dolly (L14). Last commit `5815850`.
- **2026-09-16** — Ledger created. Contract 77/77, gesture 54/54, both validators clean. README rewritten around the anatomy edition.